

SQL SALES ANALYSIS PROJECT

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TOOLS USED
MYSQL • SQL • MICROSOFT EXCEL

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Project overview

Introduction

This project focuses on analysing sales transaction data using SQL and Microsoft Excel. The objective is to extract meaningful business insights related to revenue, customer behaviour, product performance, and sales trends. Various SQL queries were used to perform data analysis and generate actionable business recommendations.

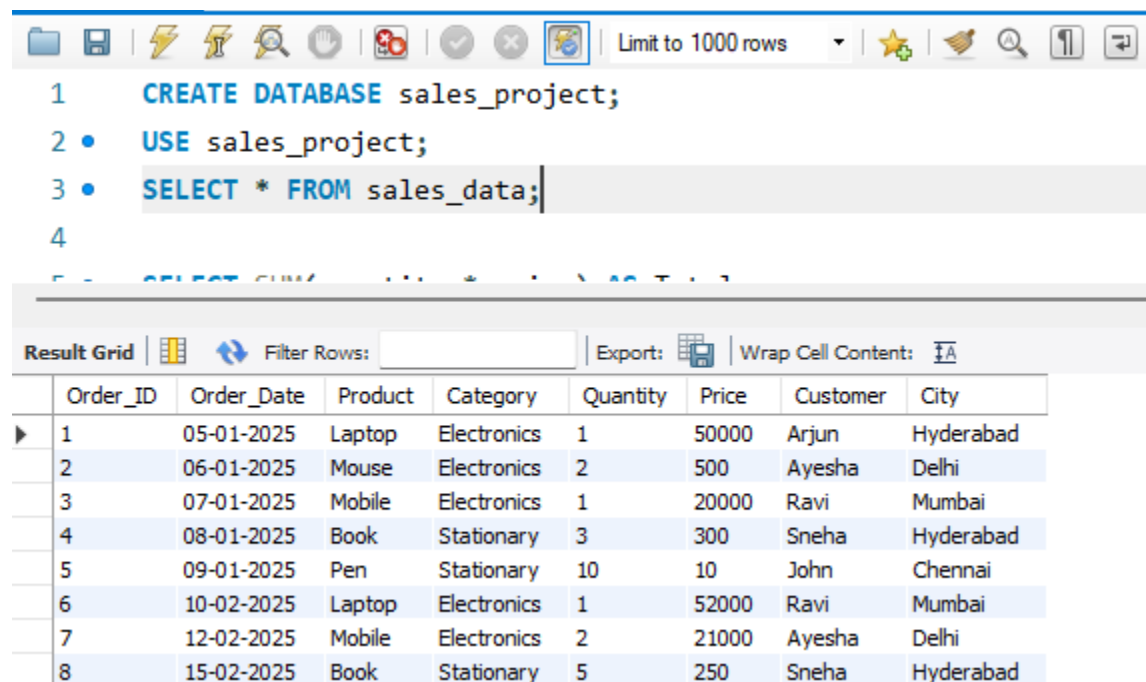
Project Objectives

- Analyse total sales revenue
- Identify top-performing products
- Analyse customer purchasing behaviour
- Compare city-wise sales performance
- Study monthly revenue trends
- Generate business insights for decision-making

Dataset Information

Column Name	Description
Order_ID	Unique order identifier
Order_Date	Date of purchase
Product	Product name
Category	Product category
Quantity	Number of units sold
Price	Price per unit
Customer	Customer name
City	Customer location

Dataset Preview



The screenshot shows a database client interface. At the top, there is a toolbar with various icons and a 'Limit to 1000 rows' dropdown. Below the toolbar, a SQL query editor contains the following queries:

```
1 CREATE DATABASE sales_project;
2 • USE sales_project;
3 • SELECT * FROM sales_data;
4
5 • SELECT SUM(Quantity) AS TotalQuantity FROM sales_data;
```

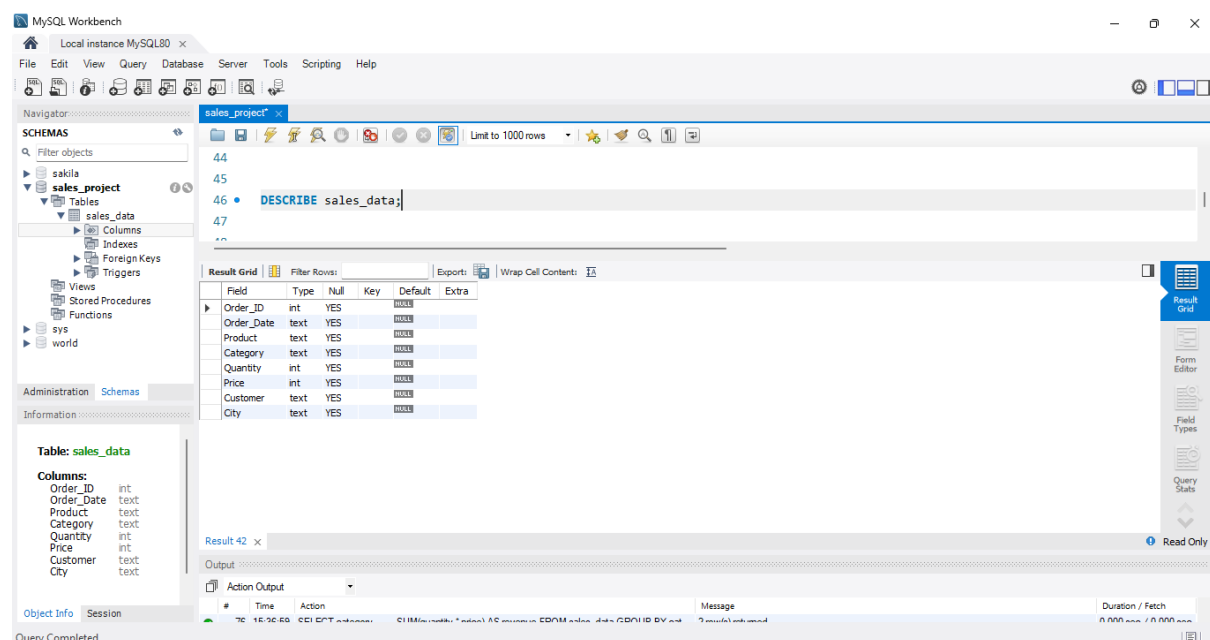
Below the query editor, a 'Result Grid' is displayed. It has a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The grid shows 8 rows of data with the following columns: Order_ID, Order_Date, Product, Category, Quantity, Price, Customer, and City.

Order_ID	Order_Date	Product	Category	Quantity	Price	Customer	City
1	05-01-2025	Laptop	Electronics	1	50000	Arjun	Hyderabad
2	06-01-2025	Mouse	Electronics	2	500	Ayesha	Delhi
3	07-01-2025	Mobile	Electronics	1	20000	Ravi	Mumbai
4	08-01-2025	Book	Stationary	3	300	Sneha	Hyderabad
5	09-01-2025	Pen	Stationary	10	10	John	Chennai
6	10-02-2025	Laptop	Electronics	1	52000	Ravi	Mumbai
7	12-02-2025	Mobile	Electronics	2	21000	Ayesha	Delhi
8	15-02-2025	Book	Stationary	5	250	Sneha	Hyderabad

Figure 1: Sample records from the sales dataset.

Database Structure

Table Structure



The screenshot shows the MySQL Workbench interface. The 'Navigator' pane on the left shows the 'sales_project' database selected. The 'Table: sales_data' is highlighted, and its structure is shown in the 'Table: sales_data' pane. The structure lists the columns and their data types:

Field	Type	Null	Key	Default	Extra
Order_ID	int	YES			
Order_Date	text	YES			
Product	text	YES			
Category	text	YES			
Quantity	int	YES			
Price	int	YES			
Customer	text	YES			
City	text	YES			

The 'Table: sales_data' pane also shows the columns and their data types in a list format:

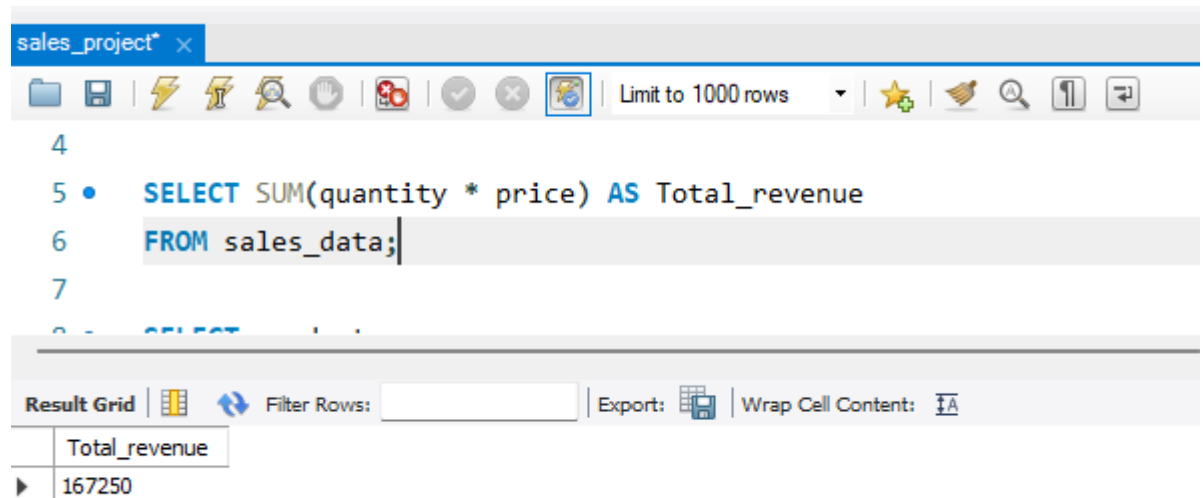
Columns:	
Order_ID	int
Order_Date	text
Product	text
Category	text
Quantity	int
Price	int
Customer	text
City	text

Figure 2: Database Schema Showing column names and data types.

Analysis Sections

1. TOTAL REVENUE ANALYSIS

Result:



The screenshot shows a SQL IDE window titled 'sales_project'. The query editor contains the following SQL code:

```
4
5 • SELECT SUM(quantity * price) AS Total_revenue
6 FROM sales_data;
7
```

Below the query editor, the 'Result Grid' tab is active, displaying a single row of results:

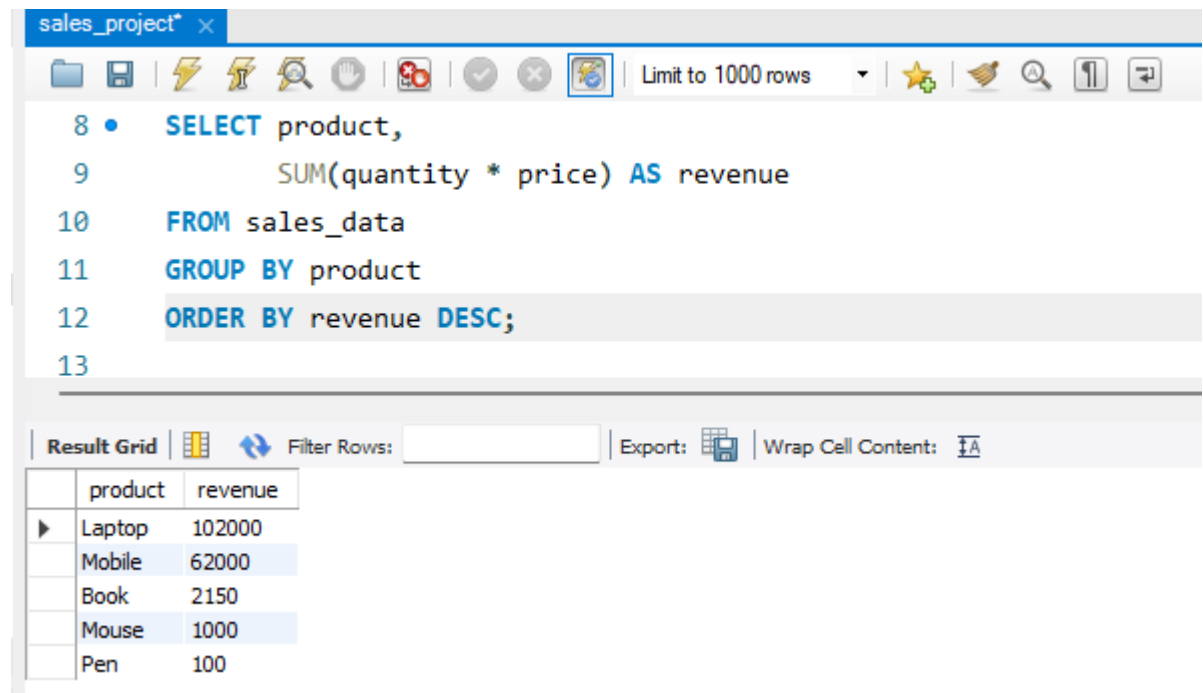
Total_revenue
167250

INSIGHT:

The business generated a total revenue of ₹167,250 during the analysis period.

2. REVENUE BY PRODUCT

Result:

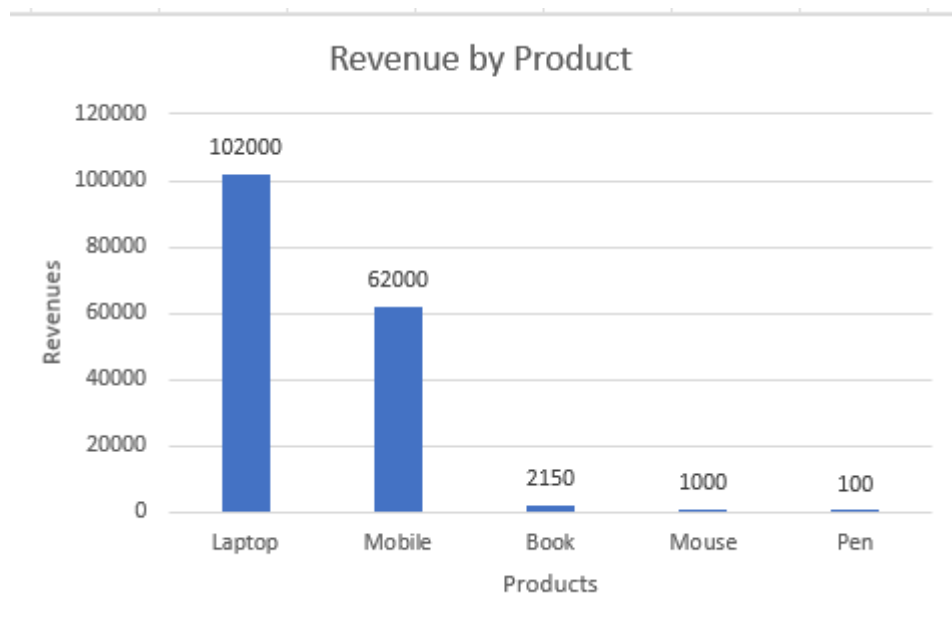


The screenshot shows a SQL IDE window titled 'sales_project'. The query editor contains the following SQL code:

```
8 • SELECT product,
9         SUM(quantity * price) AS revenue
10 FROM sales_data
11 GROUP BY product
12 ORDER BY revenue DESC;
13
```

Below the query editor, the 'Result Grid' tab is active, displaying a table of results:

product	revenue
Laptop	102000
Mobile	62000
Book	2150
Mouse	1000
Pen	100



INSIGHTS:

Laptop generated the highest revenue followed by Mobile, making Electronics the strongest revenue-generating segment.

3. CUSTOMER SPENDING ANALYSIS

Result:

sales_project* x

Limit to 1000 rows

```

21 • SELECT Customer,
22     SUM(quantity*price) AS total_money_spent
23     FROM sales_data
24     GROUP BY customer
25     ORDER BY total_money_spent DESC;
26

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Customer	total_money_spent
▶	Ravi	72000
	Arjun	50000
	Ayesha	43000
	Sneha	2150
	John	100

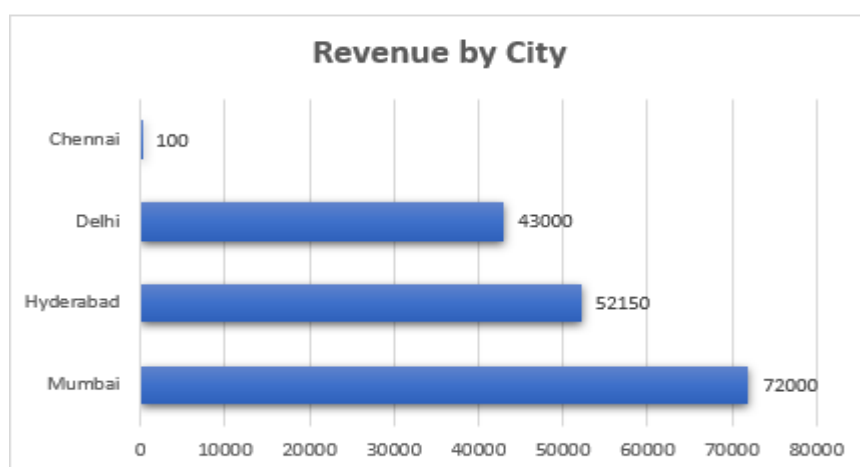
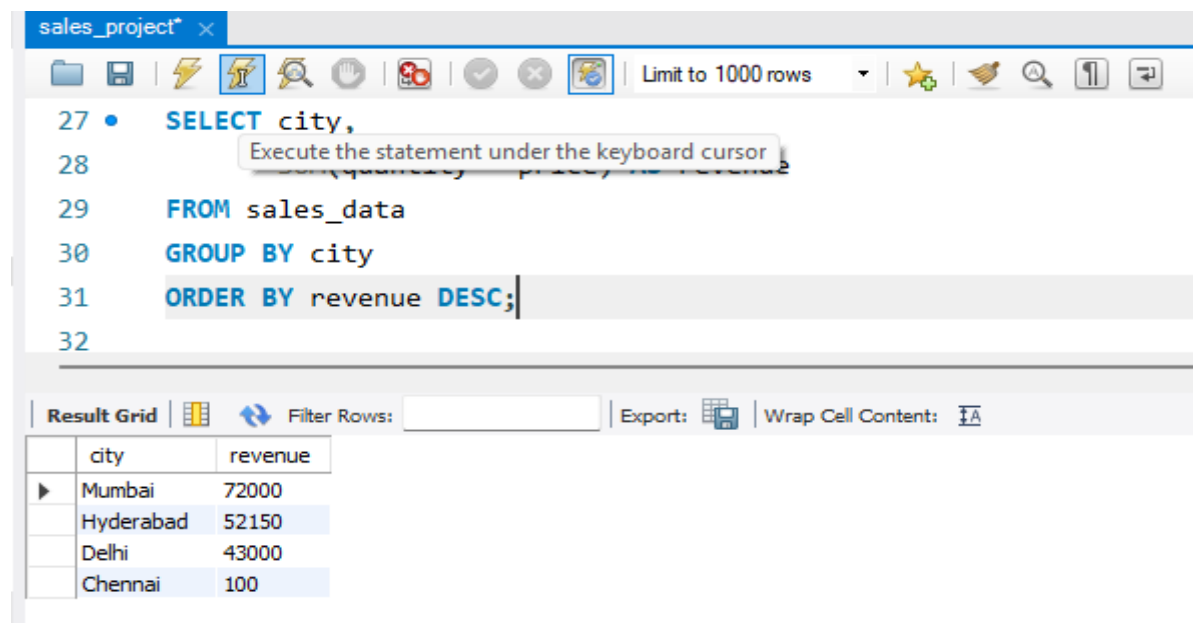


INSIGHTS:

Ravi was identified as the highest-value customer with purchases worth ₹72,000.

4. REVENUE BY CITY

Result:



INSIGHTS:

Mumbai generated the highest revenue despite having fewer orders than Hyderabad.

5. MONTHLY REVENUE ANALYSIS

Result:

sales_project x

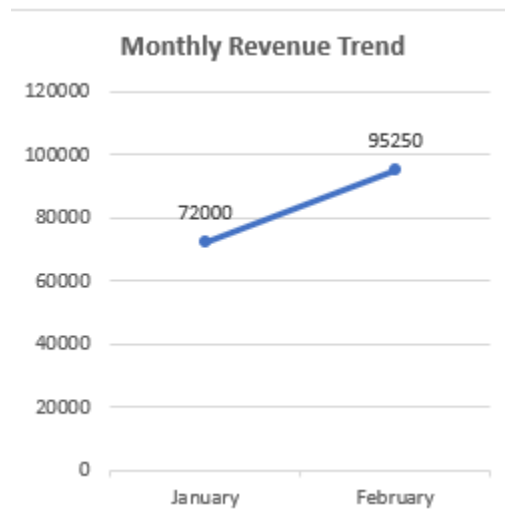
Limit to 1000 rows

```
48 • SELECT MONTH(STR_TO_DATE(Order_Date, '%d-%m-%Y')) AS month,
49      SUM(quantity * price) AS revenue
50 FROM sales_data
51 GROUP BY MONTH(STR_TO_DATE(Order_Date, '%d-%m-%Y'))
52 ORDER BY month;
53
```

Result Grid

	month	revenue
▶	1	72000
	2	95250

Filter Rows: Export: Wrap Cell Content:



INSIGHTS:

Revenue increased from ₹72,000 in January to ₹95,250 in February, indicating positive business growth.

6. CATEGORY REVENUE ANALYSIS

sales_project x

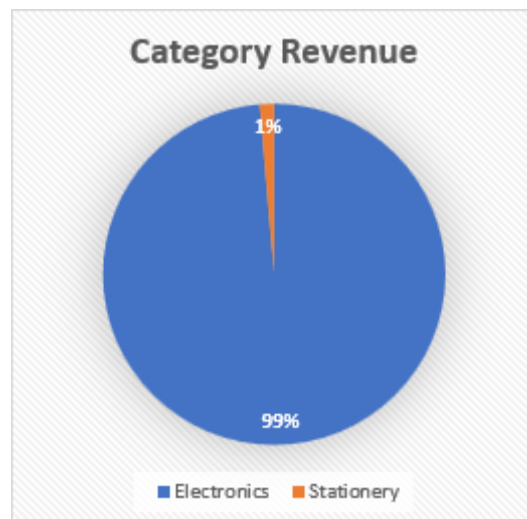
Limit to 1000 rows

```
65 • SELECT category,
66       SUM(quantity * price) AS revenue
67 FROM sales_data
68 GROUP BY category
69 ORDER BY revenue DESC;
70
```

Result Grid

	category	revenue
▶	Electronics	165000
	Stationary	2250

Export: | Wrap Cell Content: |



INSIGHTS:

Electronics accounted for the majority of business revenue while Stationery contributed a relatively small share.

Key Findings

- Total Revenue generated: ₹167,250
- Laptop was the highest revenue-generating product
- Pen was the highest-selling product by quantity
- Ravi was the top customer
- Mumbai generated the highest revenue
- Hyderabad recorded the highest number of orders
- February outperformed January in revenue
- Electronics dominated overall sales performance

Business Recommendations

1. Focus on Electronics Products

- Electronics generated the highest revenue (₹165,000). The company should prioritize inventory management and promotional activities for these products.

2. Retain High-Value Customers

- Customers such as Ravi contribute significantly to revenue. Loyalty programs and personalized offers can help improve customer retention.

3. Improve Sales in Low-Performing Cities

- Chennai recorded the lowest revenue. Targeted marketing campaigns may help increase sales in this region.

4. Monitor Monthly Revenue Trends

- Revenue increased in February compared to January. Regular sales monitoring can help identify growth opportunities and seasonal patterns.

Conclusion:

This project successfully demonstrated the use of SQL for sales data analysis and business reporting. By applying SQL queries, meaningful insights were extracted regarding customer behaviour, product performance, city-wise revenue, and sales trends. The findings can support strategic business decisions and improve overall sales performance.

Thank You

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